Data sources

JEPX spot prices
day-ahead, 9 areas, 30-min slots

JMA-MSM runs
4 channels, resolved by forecast vintage

ERA5 reanalysis
day-lagged fields and realised weather

OCCTO & imbalance
curtailment actuals, settlement prices

Weather information tiers, ordered by dissemination time

dissemination = initialisation + 2 h 30 m (JMBSC spec) · gate closure = D−1 10:00 JST

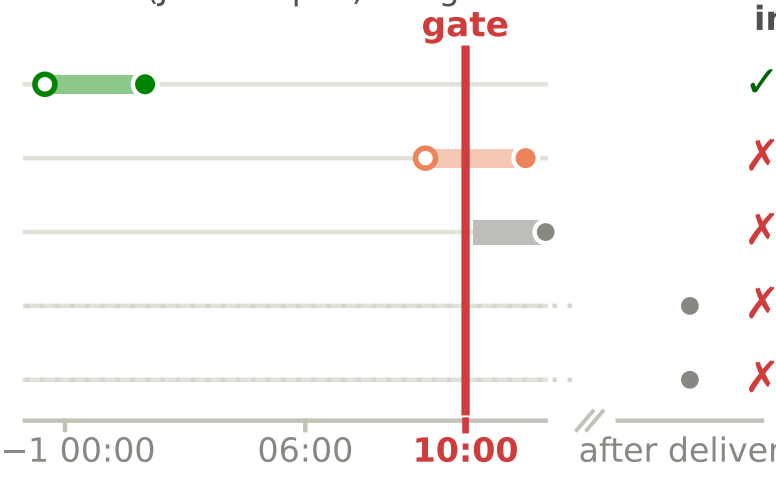
Pre-gate, run-pinned
D−2 12 UTC run

First excluded run
D−1 00 UTC run

Reconstruction
best run per timestamp

Ex-post benchmark
realised weather at delivery

Stale observation
day-lagged reanalysis



One expanding-window daily walk-forward · 714 daily walk-forward test days × 9 areas · 307,638 half-hour slots

XGBoost carries the ladder; LEAR and naive references · identical folds and one price-only baseline for every tier

A. Per-area point accuracy
ΔMAE vs price-only

+0.041 JPY/kWh
from the stale tier; pooled over 714 days, DM-significant in 7 of 9 areas

B. Vintage ladder and gate timing
five tiers, one sample

5.63× that, pre-gate
0.041→0.229 JPY/kWh; excluded run equivalent (pooled MAE, 1 tick); +0.173 over the enriched baseline

C. Commitment decision value
settled ex post at K

−169k JPY per MW-yr
547-day sample; windows −252k → +48k; the wedge governs the sign continuously — no break date is identifiable

D. Curtailment channel
floor slots vs OCCTO

19–96× more likely
near-floor prices when PV curtailment is active; 8 areas, 279,590 slots; co-occurrence, not causation